

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287102

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

JANG; Youngsoo et al.

CAMERA MODULE COMPRISING DAMPER

Abstract

An electronic device is provided. The electronic device includes a camera module. The camera module includes a camera housing, a lens assembly including at least one lens aligned along an optical axis, a focus adjustment (AF) driver configured to move the lens assembly in a direction of the optical axis, an optical image stabilization (OIS) driver configured to move the lens assembly in at least one direction intersecting the optical axis, a frame disposed in the camera housing and at least partially surrounding at least one of the AF driver or the OIS driver, and a damper including one end portion fixed to the frame or the camera housing and another end having a shape elongated toward the camera housing and configured to be bent in one direction based on the damper contacting with the camera housing.

Inventors: JANG; Youngsoo (Suwon-si, KR), PARK; Chiyong (Suwon-si, KR)

Applicant: Samsung Electronics Co., Ltd. (Suwon-si, KR)

Family ID: 91196359

Appl. No.: 19/217306

Filed: May 23, 2025

Foreign Application Priority Data

KR 10-2022-0159619

Nov. 24, 2022

KR 10-2023-0003638

Jan. 10, 2023

Related U.S. Application Data

parent WO continuation PCT/KR2023/019138 20231124 PENDING child US 19217306

Publication Classification

Int. Cl.: H04N23/68 (20230101); H04N23/51 (20230101); H04N23/54 (20230101)

U.S. Cl.:

CPC H04N23/687 (20230101); H04N23/51 (20230101); H04N23/54 (20230101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/019138, filed on Nov. 24, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0159619, filed on Nov. 24, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0003638, filed on Jan. 10, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a camera module. More particularly, the disclosure relates to a camera module including a damper for reducing noise due to contact between components.

2. Description of Related Art

[0003] With the development of digital camera manufacturing technology, an electronic device equipped with a compact and lightweight camera module has been used. Users may take advantage of various functionalities of an electronic device with camera modules, such as photographing, video recording, video call, augmented reality (AR), etc., in a convenient manner, while carrying it all the time. Typically, a camera module may include a lens assembly and an image sensor.

[0004] As it becomes more common to equip electronic devices with camera lens assemblies, efforts are now being made to downsize camera lens assemblies while also enhancing performance, such as image quality.

[0005] For example, in the AF (auto-focus) and OIS (optical image stabilization) driving systems, which operate using magnetic force, noise due to shaking may occur when the camera is in a power-off state, caused by collisions between internal components. Particularly, in some AF actuator layouts, the AF driving unit includes a portion of the OIS driving mechanism, raising concerns about the possibility of louder noise due to structural integration.

[0006] The magnet moves in the vertical direction and is coupled with a back yoke, providing attractive force and holding power based on the neutral zone of the magnet. Although this configuration may reduce some shaking noise, when a general user shakes the smartphone, the damping effect becomes insufficient, and friction noise may still be generated, potentially leading to customer complaints.

[0007] For example, auto-focus is among the techniques to allow the camera lens assembly a better performance. Autofocusing may move the lens positioned ahead of the image sensor along the optical axis depending on the distance from the subject, enabling a clear image to form on the image-forming surface of the image sensor.

[0008] Another example technology for enhancing the performance of the camera lens assembly is image stabilization. Image stabilization may also be referred to as a shake (or hand shake) function. The camera module may obtain stabilized images against disturbances by using the image stabilization function. Here, the disturbance may be the occurrence of various artifacts, such as blurring of the image obtained through the camera module due to slight shaking of the user's hand when taking a photo or video. The image stabilization function, i.e., a shake (or hand shake)

correction function, may move the lens assembly included in the camera module on a plane perpendicular to the optical axis to compensate for limited movement of the electronic device due to the user's grip or the fixing device, thereby preventing or alleviating shakes in the captured image or video.

[0009] To perform the auto-focus function and image stabilization function, the camera module may include at least one coil and magnet. The coil to which current is applied may generate electromagnetic force through electromagnetic interaction with the magnet, and the camera module may perform an auto-focus function and/or shake correction function using the generated electromagnetic force. As a method for adjusting focus or correcting shakes by electromagnetic force, various methods, such as a lens shift that moves the lens assembly, a prism shift that moves the prism, a module tilt that tilts the camera module, and an image sensor shift that moves the image sensor, may apply.

[0010] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0011] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a camera module comprising damper for reducing noise due to contact between components.

[0012] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0013] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a camera, wherein the camera includes a camera housing, a lens assembly including at least one lens aligned along an optical axis, a focus adjustment (AF) driver configured to move the lens assembly in a direction of the optical axis, an image stabilization (OIS) driver configured to move the lens assembly in at least one direction crossing the optical axis, a frame disposed in the camera housing and at least partially surrounding at least one of the AF driver or the OIS driver and a damper including one end portion fixed to the frame or the camera housing and another end portion having a shape elongated toward the camera housing and configured to be bent in one direction based on the damper contacting the camera housing.

[0014] In accordance with another aspect of the disclosure, a camera is provided. The camera includes a camera housing, a lens assembly including at least one lens aligned along an optical axis, a carrier member configured to guide the lens assembly in at least one of an optical-axis direction or a direction crossing the optical axis, and a driving member including at least one coil, and at least one magnet and a plurality of guide balls disposed to at least partially face the at least one coil, and moving the carrier member in the optical-axis direction and the direction crossing the optical axis in a ball bearing manner, wherein a frame is disposed in the camera housing and at least partially surrounding the driving member, and wherein a damper having one end portion fixed to the frame or the camera housing and another end portion having a shape elongated toward the camera housing and is configured to be bent in one direction based on the damper contacting the camera housing.

[0015] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0017] FIG. 1 is an exploded perspective view illustrating a camera module according to an embodiment of the disclosure;

[0018] FIG. 2 is a front view illustrating a camera module according to an embodiment of the disclosure;

[0019] FIG. 3 is a cross-sectional view of the camera module of FIG. 2, taken along direction A-A' according to an embodiment of the disclosure;

[0020] FIG. 4 is a cross-sectional view of the camera module of FIG. 2, taken along a direction parallel to an XY plane according to an embodiment of the disclosure;

[0021] FIG. 5A is a view illustrating a state in which a damper is disposed on a second surface of a frame according to an embodiment of the disclosure;

[0022] FIG. 5B may illustrate a placement relationship between a camera housing and a frame in a cross section of the camera module of FIG. 5A, taken along direction B-B' according to an embodiment of the disclosure;

[0023] FIG. 6A is a view illustrating a state in which a damper is disposed on an inner surface of a camera housing according to an embodiment of the disclosure;

[0024] FIG. 6B may illustrate a placement relationship between a camera housing and a frame in a cross section of the camera module of FIG. 6A, taken along direction B-B' according to an embodiment of the disclosure;

[0025] FIG. 7 is a front view illustrating a damper according to an embodiment of the disclosure;

[0026] FIG. 8 is a view illustrating a state in which the damper of FIG. 7 is bent when contacting a cover plate of a camera housing according to an embodiment of the disclosure;

[0027] FIG. 9A is a front view illustrating a damper according to an embodiment of the disclosure;

[0028] FIG. 9B is a view illustrating a state in which the damper of FIG. 9A is bent when contacting a cover plate of a camera housing according to an embodiment of the disclosure; and

[0029] FIG. 10 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

[0030] The same reference numerals are used to represent the same elements throughout the drawings.

DETAILED DESCRIPTION

[0031] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0032] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0033] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0034] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0035] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a fingerprint sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0036] FIG. 1 is an exploded perspective view illustrating a camera module according to an embodiment of the disclosure.

[0037] The camera module **10** is described below with reference to FIGS. 1 to 4, 5A, 5B, 6A, 6B, 7, 8, 9A, and 9B, but it should be noted that the category of the disclosure includes an electronic device (e.g., the electronic device **1001** of FIG. 10) as well as the camera module **10**.

[0038] In the following detailed description of FIG. 1 and the subsequent figures, the length direction of the camera module **10** may be defined as the ‘Y-axis direction’, the width direction as the ‘X-axis direction’, and/or the height direction (the thickness direction) as the ‘Z-axis direction’. In the following detailed description, the mentioned length direction, width direction, and/or height direction (or thickness direction) may indicate the length direction, the width direction, and/or the height direction (or thickness direction) of the camera module **10** or an electronic device including the camera module **10**. In an embodiment, ‘negative/positive (–/+)’ may be mentioned together with the Cartesian coordinate system exemplified in the drawings with respect to the direction in which the component is oriented. For example, referring to FIG. 1, the front surface of the camera module **10** (the front surface **122a** of the camera housing **120**) is defined as a ‘surface facing in the +Z direction (or first direction),’ and the rear surface may be defined as a ‘surface facing in the –Z direction (or second direction)’. According to an embodiment, the arrangement relationship in the height direction of a component or another component, that is, the reference as to whether a component is on/under another component, may follow the +Z-axis direction/–Z-axis direction. In other words, when a component is disposed above another component, it may mean that the component is disposed on the +Z-axis direction with respect to the other component, and when a component is disposed under another component, it may mean that the component is disposed on the –Z-axis direction with respect to the other component. Meanwhile, it should be noted that although a component is disposed on or under another component, it does not necessarily mean that the entire component is disposed on or under the entire other component. For example, it should be noted that a portion of the component may be disposed on a portion of the other component while another portion of the component is disposed under another portion of the other component. In the following description, it should be noted that when a component overlaps (or is stacked on) another component, the above description of the arrangement relationship in height direction may apply. In describing the direction, when ‘negative/positive (–/+)’ is not shown, it may be interpreted as including both the + direction and the – direction unless separately defined. For example, the ‘X-axis direction’ is interpreted as including both the +X direction and the –X direction, and the ‘Y-axis direction’ may be interpreted as including both the +Y direction and the –Y direction. In describing the direction, facing in any one of the three axes of the orthogonal coordinate system may include facing in a direction parallel to the axis. For simplicity of description, this is based on

the Cartesian coordinate system illustrated in the drawings, and it should be noted that descriptions of such directions or components do not limit various embodiments of the disclosure.

[0039] Referring to FIG. 1, the camera module **10** may include a lens assembly **110** including at least one lens, a camera housing **120**, an image sensor **130**, a first carrier **140**, a second carrier **150**, a third carrier **160**, a frame **170**, and/or a driving member **M1**, **M2**, and **M3**. Further, the camera module **10** may include at least one substrate (e.g., printed circuit board (PCB) **131**) and/or at least one flexible circuit board (FPCB) (e.g., FPCB **132**).

[0040] The camera housing **120** is a component forming the exterior of the camera module **10**, and may be provided as a cover member for providing an inner space for accommodating components inside the camera module **10** and protecting internal components. According to an embodiment, the camera housing **120** may further include a base member forming a base in addition to the components illustrated in FIG. 1. According to an embodiment, if there is a component that generates electromagnetic waves among the components accommodated in the inner space provided by the camera housing **120**, the camera housing **120** may at least partially provide an electromagnetic shielding structure. For example, the driving member **M1**, **M2**, and **M3** include a voice coil motor (VCM) that generates driving power using an electric or magnetic field, and the camera housing **120** may function as a shield can that shields electromagnetic waves generated from the voice coil motor.

[0041] The camera housing **120** may include a cover plate **122** and a side member (or side surface member) **123**. According to an embodiment, the cover plate **122** may include a front surface **122a** facing in a first direction and a rear surface **122b** facing in a second direction opposite to the first direction. Further, the cover plate **122** may include an optical opening **121** formed through the front surface **122a** and the rear surface **122b**. The optical opening **121** may provide a space for accommodating the lens assembly **110**, which is a component for moving or receiving light. According to an embodiment, the side member **123** may include a side surface **123a** connected to the cover plate **122** and facing in a third direction substantially perpendicular to the first direction and the second direction. The camera housing **120** may include a cover plate **122** and a side member **123** to provide a space for accommodating the lens assembly and/or various components. Although FIG. 1 illustrates the camera housing **120** where the side member **123** is integrally connected from the cover plate **122**, various embodiments included in the scope of the disclosure are not necessarily limited thereto, and additionally or generally, a form protruding from the base member may also be applied. As such, the camera housing **120** is not limited to any specific shape, but may have various shapes according to embodiments.

[0042] The lens assembly **110** may include at least one lens aligned along the optical axis O-I. Here, the optical axis O-I may refer to a virtual line passing through the center of the lens (centers of a plurality of lenses when there are the plurality of lenses) from the object (or external object) side O to the image side I of the image sensor. The lens assembly **110** may be at least partially accommodated in the camera housing **120**. According to an embodiment, the lens assembly **110** may be at least partially accommodated in the camera housing **120** while being surrounded by a barrel structure. The lens assembly **110** may dispose externally incident light in an appropriate number according to specifications required by the camera module **10** or the electronic device. The number of lenses included in the lens assembly **110** may not be limited to any particular embodiment. According to an embodiment, the lens assembly **110** may be disposed to partially protrude toward the outside of the camera housing **120** while being at least partially accommodated in the camera housing **120**. According to various embodiments of the disclosure, the lens assembly **110** may focus or guide light incident on the camera module **10** into the camera housing **301** while being fixedly disposed in the camera housing **120**.

[0043] The image sensor **130** may include, e.g., a sensor such as a complementary metal-oxide semiconductor (CMOS) or a charge coupled device (CCD). The image sensor **130**, however, is not limited thereto but may rather include various elements that convert light, e.g., an object image,

into an electrical image signal. The image sensor **130** may detect brightness, contrast ratio information, or color information about the subject from the light which has been transmitted through the lens assembly **110**, thereby obtaining an image for the object.

[0044] The image sensor **130** may be disposed on one surface **131a** of the PCB **131** to face the lens assembly **110**. According to an embodiment, the PCB **131** is a printed circuit board having the image sensor **130** disposed on one surface **131a** thereof, and may have a plate shape parallel to a plane (hereinafter, referred to as an “XY plane”) formed by the X-axis and the Y-axis among the spatial coordinate axes illustrated in FIG. **1**. An electrical signal may be applied to the image sensor **130** and/or a component (e.g., a coil) connected to the PCB **131** through the PCB **131**. According to an embodiment, the image sensor **130** may be connected to other component(s) (e.g., an image signal processor) through an FPCB **132** connected to the PCB **131**.

[0045] The camera module **10** may include a carrier member to guide the lens assembly **110** to reciprocate in the direction along the optical axis (+Z-axis direction or -Z-axis direction, hereinafter simply referred to as ‘D1’) and/or reciprocate in a direction (X-axis direction and/or Y-axis direction, hereinafter simply referred to as ‘D2, and/or D3’) crossing the optical axis. The carrier member may be received in the camera housing **120** and may be disposed on the image sensor **130**. While being disposed on the image sensor **130**, the lens assembly **110** may be moved in the direction D1 of the optical axis O-I by the carrier member, or additionally or alternatively, may reciprocate in the direction D2 and/or D3 crossing the optical axis. Here, the carrier member may include a first carrier **140**, a second carrier **150**, and a third carrier **160**.

[0046] The first carrier **140** may be disposed on the PCB **131**. The first carrier **140** may at least partially overlap the PCB **131**. According to an embodiment, the first carrier **140** may be disposed to at least partially surround the image sensor **130** on the surface of the first PCB **131** facing in the +Z-axis direction (the first direction). According to an embodiment, the first carrier **140** may be coupled with the PCB **131**. According to an embodiment, the first carrier **140** may be coupled with the PCB **131** to be fixed in position. The first carrier **140** may have a first opening **141** formed in a central portion thereof and may include a first plate **142** surrounding the first opening **141**.

According to an embodiment, the first opening **141** may be formed at a position corresponding to the image sensor **130** and the optical opening **121** of the camera housing **120**. According to an embodiment, the image sensor **130** and/or a filter may be positioned in the first opening **141**. For example, as the filter, an IR cut filter for blocking infrared (IR) is disposed, and the IR cut filter may overlap the image sensor **130** when disposed in the first opening **141**.

[0047] The first carrier **140** may include a first side wall **143** protruding from the first plate **142** in the +Z-axis direction (first direction). The first side wall **143** may correspond to the side member **123** of the camera housing **120** and, when the side member **123** of the camera housing **120** is formed substantially in a rectangular shape, it may include four side walls corresponding thereto. For example, the first side wall **143** includes a 1-1th side wall **143a**, a 1-2th side wall **143b**, a 1-3th side wall **143c**, and a 1-4th side wall **143d**. Coil(s) **144**, **145**, and **146** may be disposed on the first side wall **143**. The camera module **10** according to an embodiment may include coil(s) **144**, **145**, and **146** disposed on the first side wall **143** to face in a direction (e.g., X-axis and/or Y-axis direction) crossing the optical axis O-I. For example, the camera module **10** has a first coil **144** disposed on the 1-1th side wall **143a** to face in a direction (Y-axis direction) crossing the optical axis O-I, a second coil **145** disposed on the 1-2th side wall **143b** in a direction (X-axis direction) crossing the optical axis O-I, and a third coil **146** disposed on the 1-3th side wall **143c** in a direction (Y-axis direction) crossing the optical axis O-I. In the embodiment of FIG. **1**, the first coil **144** is formed as a coil forming one loop, and the second coil **145** and the third coil **146** are provided as coils forming two loops, but the disclosure is not limited thereto. Another electronic component(s) (e.g., sensors **147**, **148**, and **149**) may be disposed inside the coil(s) **144**, **145**, and **146**. The coil(s) **144**, **145**, and **146** and other electronic component(s) are described below in detail with reference to FIGS. **2** to **4**.

[0048] The second carrier **150** may be disposed on the first carrier **140**. According to an embodiment, the second carrier **150** may be disposed on a surface **142a** of the first carrier **140** facing in the +Z-axis direction (first direction). The second carrier **150** may at least partially overlap the first carrier **140**. The second carrier **150** may include a second plate **152** having a second opening **151** formed in a central portion thereof and surrounding the second opening **151**. According to an embodiment, the second opening **151** may be formed at a position corresponding to the first opening **141**. According to an embodiment, unlike the first carrier **140** fixedly coupled to the PCB **131**, the second carrier **150** is not fixedly coupled to the first carrier **140** but may be coupled so that the position between the first carrier **140** and the counterpart is changeable. As is described below, the second carrier **150** is configured to be movable in the optical axis O-I direction **D1**, which is related to an auto-focus AF operation, and thus it may be referred to as an 'AF carrier'.

[0049] The second carrier **150** may include a second side wall **153** extending from the second plate **152**. The second side wall **153** may face at least a portion of the first side wall **143**. The second carrier **150** may reciprocate along the optical axis O-I direction **D1** while the second side wall **153** faces at least a portion of the first side wall **143**. As a first rail structure is provided between the second carrier **150** and the first carrier **140**, the linear movement of the second carrier **150** in the optical axis O-I direction **D1** may be guided and the second carrier **150** may be restricted from moving in another direction with respect to the first carrier **140**. According to an embodiment, the first rail structure may be implemented by forming a rail groove(s) extending parallel to the optical axis (O-I) direction in the inner wall of the first side wall **143** of the first carrier **140**, forming a rail groove(s) extending parallel to the optical axis (O-I) direction in the outer wall of the second side wall **153** of the second carrier **150**, and arranging a plurality of first guide balls **180** between the rail groove(s) of the first side wall **143** and the rail groove(s) of the second side wall **153**. Although FIG. **1** illustrates that the first guide balls **180** are disposed in two different areas, the disclosure is not limited thereto, and the first guide balls may be disposed in one area or three or more different areas. The rail groove(s) formed in the inner wall of the first side wall **143** of the first carrier **140** and the outer wall of the second side wall **153** of the second carrier **150** are rails for guiding the first guide ball **180**, and may be formed in, e.g., a V-groove shape.

[0050] According to an embodiment, the camera module **10** may include a first magnet **154** disposed to face in the direction (Y-axis direction) crossing the optical axis O-I at a position corresponding to the 1-1th side wall **143a** of the first carrier **140** of the second side wall **153** of the second carrier **150**. The first magnet **154** and the first coil **144** may constitute the first driver **M1**. According to an embodiment, the first driver **M1** may drive the lens assembly **110** in a ball bearing manner using the first guide ball **180**. The first driver **M1** is described in detail with reference to FIGS. **2** to **4**.

[0051] The third carrier **160** may be disposed on the second carrier **150**. According to an embodiment, the third carrier **160** may be disposed on a surface **152a** of the second carrier **150** facing in the +Z-axis direction (first direction). The third carrier **160** may at least partially overlap the second carrier **150**. A third opening **161** may be formed in the central portion of the third carrier **160**. The third opening **161** may be formed at a position corresponding to the second opening **151**. According to an embodiment, the third carrier **160** may not be fixedly coupled to the second carrier **150**, but may be coupled so that the relative positions thereof may be changed. The third carrier **160** may linearly reciprocate in a direction **D2** and/or **D3** crossing the optical axis while being disposed on the second carrier **150**. Since the movement in the direction **D2** and/or **D3** crossing the optical axis is related to an optical image stabilization (OIS) operation, the third carrier **160** may be referred to as an 'OIS carrier'.

[0052] The third carrier **160** may include a 3-1th carrier **160a** and a 3-2th carrier **160b** capable of linearly reciprocating in two different directions **D2** or **D3**, respectively, crossing the optical axis. For example, if the 3-2th carrier **160a** linearly reciprocates in the direction **D2**, the 3-1th carrier

160b linearly reciprocates in the direction **D3**, whereas if the 3-1th carrier **160a** linearly reciprocates in the direction **D3**, the 3-2th carrier **160b** may linearly reciprocate in the direction **D2**. In the following description, an embodiment where the 3-1th carrier **160a** linearly reciprocates in the **D2** direction and the 3-2th carrier **160b** linearly reciprocates in the **D3** direction is mainly described.

[0053] As the second rail structure is provided between the 3-1th carrier **160a** and the second carrier **150**, a linear movement of the 3-1th carrier **160a** in the direction **D2** crossing the optical axis **O-I** may be guided and the third carrier **160** may be restricted from moving in another direction with respect to the second carrier **150**. According to an embodiment, the second rail structure may be implemented by forming the rail groove(s) extending in the direction **D2** crossing the optical axis **O-I** in the bottom surface (the surface facing in the $-Z$ -axis direction) of the 3-1th carrier **160a**, forming the rail groove(s) **152b** extending in the direction **D2** crossing the optical axis **O-I** in the surface **152a** facing in the $+Z$ -axis direction of the second carrier **150**, and arranging a plurality of second guide balls **190** between the rail groove(s) in the bottom surface of the 3-1th carrier **160a** and the rail groove(s) **152b** of the second plate **152**. Although FIG. 1 illustrates that the second guide balls **190** are disposed in four different areas, the disclosure is not limited thereto, and the second guide balls may be disposed in three or less different areas, or five or more different areas. The rail groove(s) of the bottom surface of the 3-1th carrier **160a** and the rail groove(s) **152b** of the second plate **152** are rails for guiding the second guide ball **190**, and may be, e.g., formed in a V-groove shape.

[0054] As the third rail structure is provided between the 3-2th carrier **160b** and the 3-1th carrier **160a**, a linear movement of the 3-2th carrier **160b** in the direction **D3** crossing the optical axis **O-I** may be guided, and movement of the 3-2th carrier **160b** in a different direction with respect to the 3-1th carrier **160a** may be restricted. The third rail structure may be orthogonal to the second rail structure. According to an embodiment, the third rail structure may be implemented by forming the rail groove(s) extending in the direction **D3** crossing the optical axis **O-I** in the bottom surface (the surface facing in the $-Z$ -axis direction) of the 3-2th carrier **160b**, forming the rail groove(s) extending in the direction **D3** crossing the optical axis **O-I** in the upper surface (the surface facing in the $+Z$ -axis direction) of the 3-1th carrier **160a**, and arranging a plurality of second guide balls **190** between the rail groove(s) in the bottom surface of the 3-2th carrier **160b** and the rail groove(s) in the upper surface of the 3-1th carrier **160a**. The rail groove(s) in the bottom surface of the 3-2th carrier **160b** and the rail groove(s) of the 3-1th carrier **160a** are also rails for guiding the second guide ball **190** and may be formed in, e.g., a V-groove shape.

[0055] Magnet(s) **165** and **166** may be disposed around the third carrier **160**. According to an embodiment, the camera module **10** may include a second magnet **165** disposed to face in the direction (X -axis direction) crossing the optical axis **O-I** at a position corresponding to the 1-2th side wall **143b** of the first carrier **140**, around the third carrier **160**, and a third magnet **166** disposed to face in a direction (Y -axis direction) crossing the optical axis **O-I** at a position corresponding to the 1-3th side wall **143c** of the first carrier **140**, around the third carrier **160**. According to an embodiment, the second magnet **165**, together with the second coil **145**, may constitute a second driver **M2**. The third magnet **166**, together with the third coil **146**, may constitute the third driver **M3**. According to an embodiment, the second driver **M2** and the third driver **M3** may drive the lens assembly **110** in a ball bearing manner using the second guide ball **190**. The second driver **M2** and the third driver **M3** are described below in detail with reference to FIGS. 2 to 4.

[0056] The frame **170** may be disposed on the third carrier **160**. According to an embodiment, the frame **170** may be disposed on an upper surface (a surface facing in the $+Z$ -axis direction) of the third carrier **160**. The frame **170** may at least partially overlap the third carrier **160**. According to an embodiment, the frame **170** may be fixedly coupled to the second carrier **150**. According to an embodiment, the frame **170** is disposed inside the camera housing, and itself may at least partially surround the first driver (or focus adjustment driver) **M1** and/or the second and third drivers **M2**

and M3 (or image stabilization driver). According to an embodiment, the frame **170** may be referred to as an 'AF carrier cover' in terms of being fixedly coupled to the second carrier **150** and disposed to surround the first driver M1. Additionally or alternatively, the frame **170** may be referred to as an 'OIS carrier cover' in terms of being disposed on the third carrier **160** and also surrounding the second and third drivers M2 and M3. According to an embodiment, the frame **170** may be provided to cover the carrier assembly after forming the carrier assembly where the second carrier **150** and the third carrier **160** and internal components (e.g., coils, magnets, guide balls) are assembled. The frame **170** may include a fourth plate **172** having a fourth opening **171** formed in the central portion and surrounding the fourth opening **171**. The frame **170** may include a fourth side wall **173** protruding from the fourth plate **172** in the -Z-axis direction (the second direction). According to an embodiment, the fourth side wall **173** may be formed to substantially surround the carrier assembly. Alternatively, the fourth side wall **173** may have a shape where a frame piece extending from the fourth plate **172** is folded to protrude in the -Z-axis direction (second direction) as illustrated in FIG. 1. According to an embodiment, the frame **170** may be formed of a metal material and/or a non-metal material (e.g., polymer), and may be formed of a rigid material to protect the carrier assembly from external impact.

[0057] The camera module **10** may include a plurality of side wall(s) **173**. At least some of the plurality of side wall(s) **173** may have an opening-shaped fastening structure as illustrated in the drawings, which may be coupled to a hook-shaped fastening structure (not shown) formed on the second side wall **153** of the second carrier **150**. However, the disclosure is not limited thereto, and a hook-shaped fastening structure may be provided on the plurality of side walls(s) **173**, and an opening-shaped fastening structure may be provided on the second side wall **153** of the second carrier **150** corresponding thereto. According to an embodiment of the disclosure, a damper **200** may be provided in at least one of a plurality of side wall(s) **173**. One end portion of the damper **200** may be fixed on the second surface **173a** of the plurality of side walls **173** of the frame **170**, and the other end thereof may extend toward the camera housing **120**. The camera module and the electronic device including the same of the disclosure may distribute the impact applied to the camera module **10** and/or reduce noise using the damper **200** provided on the second surface **173a** of the frame **170**. Hereinafter, the damper **200** is described in more detail with reference to FIGS. 5A, 5B, 6A, 6B, 7, 8, 9A, and 9B.

[0058] FIG. 2 is a front view illustrating a camera module according to an embodiment of the disclosure.

[0059] FIG. 3 is a cross-sectional view of the camera module of FIG. 2, taken along direction A-A' according to an embodiment of the disclosure.

[0060] FIG. 4 is a cross-sectional view of the camera module of FIG. 2, taken along a direction parallel to the XY plane according to an embodiment of the disclosure.

[0061] Referring to FIGS. 2 to 4, behavior during focus adjustment and behavior during image stabilization by the driving member M1, M2, and M3 of the camera module **10** according to an embodiment of the disclosure may be described.

[0062] Referring to FIGS. 1 to 4, the driving member M1, M2, and M3 may include at least one coil and at least one magnet disposed to partially face the at least one coil. In this case, the at least one magnet may be disposed to face at least one coil in a state of facing in the direction (e.g., X-axis direction and/or Y-axis direction) crossing the optical axis or be disposed to face at least one coil in a state of facing in the optical axis direction (e.g., Z-axis direction). A plurality of driving member M1, M2, and M3 may be provided. According to an embodiment, the driving member M1, M2, and M3 may include a first driver M1, a second driver M2, and a third driver M3. In this case, the first driver M1 may be a focus adjustment driver (AF) for automatically adjusting the focus of the camera module **10**, and the second driver M2 and the third driver M3 may be image stabilization drivers (OIS drivers) that are drivers for optical image stabilization (OIS) of the camera module **10**. The driving members M1, M2, and M3 may allow the lens assembly **110** to

move independently in three different directions (e.g., three axial directions of the spatial coordinate system of FIG. 1). According to an embodiment, various embodiments of the disclosure may be applied to the lens shift-type camera module **10** as described above. However, the disclosure is not necessarily limited thereto, and when the focus adjustment driver (AF driver) includes a voice coil motor (VCM), it may be applied to the camera module **10** of a type (e.g., image sensor shift) other than the lens shift type.

[0063] The first driver M1 may include a first coil **144** disposed in the first carrier **140**, and a first magnet **154** disposed at a position corresponding to the first coil **144** on the second side wall **153** of the second carrier **150**. According to an embodiment, the camera module **10** may further include a first supporting member **154a** connected to the second carrier **150** and supporting the first magnet **154**, and the first magnet **154** may face the first coil **144** in a direction (Y-axis direction) crossing the optical axis O-I while being disposed on the first supporting member **154a**. The first carrier **140** may be fixed and disposed on the PCB **131**. Accordingly, the position of the first coil **144** may also be fixed. If a current is applied to the first coil **144** whose position is fixed, the first magnet **154** and/or the second carrier **150** where the first magnet **154** is disposed may be driven in the optical axis O-I direction D1 by the first magnet **154** and the magnetic field applied therearound. As described above in FIG. 1, rail groove(s) may be formed in the respective side walls (the first side wall **143** and the second side wall **153**) of the first carrier **140** and the second carrier **150**. For example, rail groove(s) **143-1** are formed in the inner wall of the first side wall **143**, and rail groove(s) **153-1** may be formed in the outer wall of the second side wall **153**. Further, a first guide ball **180** may be disposed between the rail groove(s) **143-1** formed in the inner wall of the first side wall **143** and the rail groove(s) **153-1** formed in the outer wall of the second side wall **153**. A predetermined gap may be formed between the second carrier **150** and the first carrier **140** so that the second carrier **150** does not interfere with the first carrier **140**. However, since the second side wall **153** of the second carrier **150** is surrounded by the first side wall **143** of the first carrier **140**, the movement of the second carrier **150** in at least some directions may be limited by the first carrier **140**. If a current is applied to the first coil **144**, the second carrier **150** may move in the direction D1 of the optical axis O-I guided by the rail groove(s) **143-1** and **153-1** and the first guide ball **180**, but may be restricted from moving in other directions. If the direction of the current applied to the first coil **144** is changed, the driving direction of the first magnet **154** and the second carrier **150** where the first magnet **154** is disposed may be changed. The lens assembly **110** may move depending on the behavior of the second carrier **150**, and accordingly, a focus adjustment (AF) operation may be implemented.

[0064] The second driver M2 may include a second coil **145** disposed in the first carrier **140**, and a second magnet **165** disposed at a position corresponding to the second coil **145** around the third carrier **160**. According to an embodiment, the camera module **10** may further include a second supporting member **165a** connected to the 3-1th carrier **160a** and supporting the second magnet **165**, and the second magnet **165** may face the second coil **145** in a direction (X-axis direction) crossing the optical axis O-I while being disposed on the second supporting member **165a**. The 3-1th carrier **160a** may move together with the second magnet **165** through the second supporting member **165a**. If a current is applied to the second coil **145** whose position is fixed, the second magnet **165** and the 3-1th carrier **160a** may be driven in a direction D2 crossing the optical axis O-I by the second magnet **165** and magnetic field applied therearound. As described above with reference to FIG. 1, rail groove(s) may be formed between the second carrier **150** and the 3-1th carrier **160a**. For example, rail groove(s) **152b** are formed in the upper surface (the surface facing in the +Z axis) **152a** of the second plate **152** of the second carrier **150**, and rail groove(s) corresponding thereto may be formed in the bottom surface of the 3-1th carrier **160a**. Further, a second guide ball **190** may be arranged between the rail groove(s) **152b** formed in the upper surface **152a** of the second carrier **150** and the rail groove(s) formed in the bottom surface of the 3-1th carrier **160a**. According to an embodiment, a predetermined gap may be formed between the 3-1th

carrier **160a** and the second carrier **150** so that the 3-1th carrier **160a** does not interfere with the second carrier **150**. However, since the 3-1th carrier **160a** is surrounded by the second side wall **153** of the second carrier **150**, the movement of the 3-1th carrier **160a** may be at least partially limited. If a current is applied to the second coil **145**, the 3-1th carrier **160a** may move in the direction **D2** crossing the optical axis **O-I** guided by the second guide ball **190** and the rail groove(s) between the 3-1th carrier **160a** and the second carrier **150**, but its movement in other directions may be limited. If the direction of the current applied to the second coil **145** is changed, the driving direction of the second magnet **165** and the 3-1th carrier **160a** may be changed. The 3-2th carrier **160b** may move, depending on the behavior of the 3-1th carrier **160a**. The lens assembly **110** may also move depending on the behavior of the 3-1th carrier **160a**, and accordingly, the first image stabilization **OIS1** operation may be implemented.

[0065] The third driver **M3** may include a third coil **146** disposed in the first carrier **140**, and a third magnet **166** disposed at a position corresponding to the third coil **146** around the third carrier **160**. According to an embodiment, the camera module **10** may further include a third supporting member **166a** connected to the third carrier **160** and supporting the third magnet **166**, and the third magnet **166** may face the third coil **146** in a direction (**X**-axis direction) crossing the optical axis **O-I** while being disposed on the third supporting member **166a**. If a current is applied to the third coil **146** whose position is fixed, the third magnet **166** and/or the third carrier **160** where the third magnet **166** is disposed may be driven in the direction **D3** crossing the optical axis **O-I** by the third magnet **166** and the magnetic field applied therearound. As described above with reference to FIG. 1, rail groove(s) may be formed between the 3-2th carrier **160b** and the 3-1th carrier **160a**. For example, a rail groove(s) are formed in the bottom surface of the 3-2th carrier **160b**, and a rail groove(s) corresponding thereto may be formed in the upper surface of the 3-1th carrier **160a**. Further, a second guide ball **190** may be arranged between the rail groove(s) formed in the bottom surface of the 3-2th carrier **160b** and the rail groove(s) formed in the upper surface of the 3-1th carrier **160a**. According to an embodiment, a predetermined gap may be formed between the 3-2th carrier **160b** and the 3-1th carrier **160a** so that the 3-2th carrier **160b** does not interfere with the 3-1th carrier **160a**. Further, a predetermined gap may be formed between the 3-2th carrier **160b** and the frame so that the 3-2th carrier **160b** does not interfere with the frame **170**. Since the 3-2th carrier **160b** is surrounded by the 3-1th carrier **160a** and the frame **170**, movement of the 3-2th carrier **160b** may be at least partially limited. If a current is applied to the third coil **146**, the 3-2th carrier **160b** may move in the direction **D3** crossing the optical axis **O-I** guided by the second guide ball **190** and the rail groove(s) between the 3-2th carrier **160b** and the 3-1th carrier **160a**, but its movement in other directions may be limited. If the direction of the current applied to the third coil **146** is changed, the driving direction of the third magnet **166** and the third carrier **160** where the second magnet **165** is disposed may be changed. The lens assembly **110** may also move depending on the behavior of the third carrier **160**, and accordingly, the second image stabilization **OIS2** operation may be implemented.

[0066] In the embodiments of FIGS. 1 to 4, an embodiment in which the first driver **M1** adopts a Lorenz-type VCM to which Faraday's left-hand law is applied, and the second driver **M2** and the third driver **M3** adopt a solenoid-type VCM to which Ampere's right-hand law is applied is disclosed. However, it is not necessarily limited thereto, and it should be noted that an embodiment in which a solenoid-type VCM is applied to the first driver **M1** and a Lorenz-type VCM is applied to the second driver **M2** and the third driver **M3** may also be included in the embodiments of the disclosure.

[0067] According to an embodiment, the electronic device (e.g., the electronic device **1001** of FIG. 10) or the processor (e.g., the processor **1020** of FIG. 10) may move the lens assembly **110** of the camera module **10** in the optical axis **O-I** direction **D1** based on phase difference information or contrast information about an image or video. In this case, the first driver **M1** may perform an operation of moving the lens assembly **110** in the optical axis **O-I** direction **D1**.

[0068] According to an embodiment, the electronic device (e.g., the electronic device **1001** of FIG. **10**) or the sensor module (e.g., a gyro sensor and/or an acceleration sensor as one of the sensor modules **1076** of FIG. **10**) may be configured to detect vibration applied to the housing or camera module **10** of the electronic device (e.g., the electronic device **1001** of FIG. **10**). The electronic device (e.g., **1001** of FIG. **10**) or the processor (e.g., the processor **1020** of FIG. **10**) may move the lens assembly **110** in at least one of directions **D2** or **D3** crossing the optical axis O-I based on the vibration detected through the sensor module **1076**. According to an embodiment, the electronic device or processor may identify whether the vibration detected by the sensor module **1076** is the user's handshake and, when it is identified as the user's handshake, may move the lens assembly **110**. Identifying vibration in a general use environment and identifying hand shake in photographing may be based on data on the acceleration or direction of vibration and/or vibration force, and such data may be stored in the electronic device and/or memory (e.g., the memory **1030** of FIG. **10**). By moving the lens assembly **110** in the direction crossing the optical axis, it is possible to compensate for the vibration applied to the housing or camera module **10** of the electronic device (e.g., the electronic device **1001** of FIG. **10**) by an external force and increase the quality of the captured image or video.

[0069] According to an embodiment, the camera module **10** is an example of the sensor module (e.g., the sensor module **1076** of FIG. **10**), and may further include a sensor **147** for detecting the displacement of the first magnet **154** during the focus adjustment AF operation. According to an embodiment, the sensor **147** may be disposed in a hollow surrounded by the first coil **144**. Additionally, the camera module **10** may further include a sensor(s) **148** and **149** for detecting displacement of the second magnet **165** and/or the third magnet **166** during the image stabilization operation as another example of the sensor module (e.g., the sensor module **1076** of FIG. **10**). According to an embodiment, the sensor(s) **148** and **149** may be disposed in the hollows surrounded by the second coil **145** and the third coil **146**, respectively.

[0070] According to an embodiment, the camera module **10** may include at least one yoke. At least one yoke may align, e.g., the electric field and/or the magnetic field generated from the driving member **M1**, **M2**, and **M3** within a designated area or space. For example, at least one yoke contributes to reducing the power applied to the coil(s) **144**, **145**, and **146** or downsizing the driving members **M1**, **M2**, and **M3** by allowing the electric and/or magnetic fields generated from the driving members **M1**, **M2**, and **M3** to act within the designated area or space. According to an embodiment, the camera module **10** may be easily mounted in a compact electronic device by reducing power consumption or downsizing the driving member **M1**, **M2**, and **M3**, and may increase power efficiency in an auto-focus operation or an image stabilization operation. Referring to FIG. **3**, the first driver **M1** may include a first yoke **Y**, and the first yoke **Y** may be positioned opposite to the first magnet **154** with respect to the position of the first coil **144**. Further, the first yoke **Y** may serve to increase the strength of the magnetic force acting between the first coil **144** and the first magnet **154**. According to an embodiment, although not separately illustrated in the drawings, similar to the arrangement and operation principle of the first yoke **Y**, the second driver **M2** may include a second yoke, and/or the third driver **M3** may include a third yoke.

[0071] Referring to FIG. **3**, when the first magnet **154** linearly reciprocates in the optical axis O-I direction, a suction force by the first yoke **Y** may be provided, and a holding force of maintaining the posture with respect to the neutral zone by the magnetic field of the first magnet **154** may be provided. As a result, when the electronic device (e.g., the electronic device **1001** of FIG. **10**) or the camera module **10** is slightly shaken, shaking noise due to the shake may be partially reduced, but when the electronic device is severely shaken while the user is on the move or is using the electronic device (e.g., the electronic device **1001** of FIG. **10**), significant shaking noise may occur.

[0072] According to an embodiment, a damper having a flat plate structure may be formed on a first surface **172a** of the fourth plate **172** of the frame **170**, but the damper having the flat plate structure is a structure that is not enough to absorb and/or distribute an impact, and may not

sufficiently reduce noise.

[0073] In the disclosure, as illustrated in FIG. 1, there may be provided a camera module **10** capable of significantly reducing noise and facilitating absorption or distribution of impact by including a damper **200** having one end portion fixed to the frame and the other end extending toward the camera housing and, when contacting the camera housing, bent in one direction.

[0074] According to an embodiment, the damper **200** is positioned where the driver (e.g., the first driver M1) for focus adjustment (AF) and the drivers (e.g., the second driver M2 and the third driver M3) for image stabilization (OIS) are not disposed, not to affect the focus adjustment (AF) operation and image stabilization (OIS) operation. Referring to FIGS. 1 to 4, the driver (e.g., the first driver M1) for the focus adjustment (AF) may be disposed at a position corresponding to the 1-1th side wall **143a**, and the drivers (e.g., the second driver M2 and the third driver M3) for the image stabilization (OIS) may be disposed at positions corresponding to the 1-2th side wall **143b** and the 1-3th side wall **143c**. Accordingly, the damper **200** may be disposed at a position corresponding to the 1-4th side wall **144c** except for the 1-1th side wall **143a**, the 1-2th side wall **143b**, and the 1-3th side wall **143c** of the first carrier **140**.

[0075] FIG. 5A is a view illustrating a state in which a damper is disposed on a second surface of a frame of a camera module according to an embodiment of the disclosure.

[0076] FIG. 5B may illustrate a placement relationship between a camera housing and a frame in a cross section of the camera module of FIG. 5A, taken along direction B-B' according to an embodiment of the disclosure.

[0077] Referring to FIGS. 5A and 5B, the camera module **10** of the disclosure may include a damper **200** disposed on the frame **170**.

[0078] When the electronic device (e.g., the electronic device **1001** of FIG. 10) and/or the camera module **10** are shaken, the frame **170** may move closer to or away from the camera housing **120** along the first direction (+Z direction). While the electronic device (e.g., the electronic device **1001** of FIG. 10) and/or the camera module **10** are not shaken, the frame **170** and the camera housing **120** may be spaced apart from each other by the first gap g1. Here, the state in which the electronic device (e.g., the electronic device **1001** of FIG. 10) and/or the camera module **10** are not shaken may mean an initial state in which the damper **200**, which is described below, is not crushed or deformed. Further, the state in which the electronic device (e.g., electronic device **1001** of FIG. 10) and/or the camera module **10** is not shaken may include a state in which power is not applied to the electronic device (e.g., electronic device **1001** of FIG. 10) and/or the camera module **10**, i.e., a non-powered state. The damper **200** of the disclosure may be formed to be spaced apart from the camera housing **120** by the second gap g2 in a state in which the electronic device (e.g., the electronic device **1001** of FIG. 10) and/or the camera module **10** is not shaken. When the electronic device is shaken so that the frame **170** approaches the camera housing **120**, the damper **200** fixed to the frame **170** may also move in a direction of approaching the camera housing **120**, and when the frame **170** further approaches the camera housing **120**, the damper **200** may contact the rear surface **122b** of the camera housing **120**. The damper **200** in contact with the rear surface **122b** of the camera housing **120** may be pressed by the camera housing **120** and bent in one direction as illustrated in FIG. 8 to be described below.

[0079] The frame **170** may include a first surface **172a** facing the cover plate **122** of the camera housing **120** and a second surface **173a** facing the side member **123** of the camera housing **120**. Further, the damper **200** may be disposed on the second surface **173a** of the frame **170**. For example, when the damper **200** is disposed on the first surface **172a** of the frame **170**, it is difficult to secure a height (a height in the Z-axis direction) of the damper **200** in the optical axis O-I direction. Therefore, when the damper **200** is disposed on the first surface **172a** of the frame **170**, it may be difficult to absorb and/or distribute the impact generated when the frame **170** and the camera housing **120** collide through the damper **200**. Further, when the damper **200** is disposed on the first surface **172a** of the frame **170**, the noise generated when the frame **170** and the camera

housing **120** collide may be large. In the disclosure, as the damper **200** is disposed on the second surface **173a** of the frame **170**, it may be easy to secure a height of the damper **200** in the optical axis (O-I) direction, and accordingly, absorb and/or disperse the impact generated when the frame **170** and the camera housing **120** collide through the damper **200** while reducing noise.

[0080] The damper **200** may include at least a portion **210** fixed to the frame **170** and a portion **220** extending inclined with respect to the optical axis, and the extending portion **220** may be disposed in the space **S1** between the frame **170** and the side member **123**. According to an embodiment, the damper **200** may be disposed in the space **S1** between the fourth side wall **173** of the frame **170** and the side member **123** of the camera housing **120**. Since a predetermined gap is formed between the fourth side wall **173** and the side member **123** so as not to interfere with the camera housing **120**, the gap may be used as a space **S1** where the damper **200** may be disposed. According to an embodiment, the space **S1** between the fourth side wall **173** of the frame **170** and the side member **123** of the camera housing **120** may correspond to a thickness of the first side wall **143** of the first carrier **140**. According to an embodiment, the 1-4th side wall **143d** of the first carrier **140** may include a recess **143d-1** formed so as not to interfere with the damper **200** while the damper **200** reciprocates in the optical axis O-I direction **D1** during the focus adjustment (AF) operation.

[0081] The damper **200** may include a first portion **210** and a second portion **220** extending from the first portion **210** to be a single body. The first portion **210** may be a portion of the damper **200** whose position and/or shape is fixed to the second surface **173a** of the frame **170**, and the second portion **220** may be a portion whose position and/or shape is variable. According to an embodiment, the first portion **210** and the frame **170** may be formed through a double injection-molding process. According to an embodiment, the first portion **210** may be formed in a hook shape and may be formed by being press-fitted into the opening **174** formed in the fourth side wall **173** of the frame **170**.

[0082] FIG. **6A** is a view illustrating a state in which a damper is disposed on an inner surface of a camera housing according to an embodiment of the disclosure.

[0083] FIG. **6B** may illustrate a placement relationship between a camera housing and a frame in a cross section of the camera module of FIG. **6A**, taken along direction B-B' according to an embodiment of the disclosure.

[0084] Referring to FIGS. **6A** and **6B**, a camera module **10** of the disclosure may include a damper **300** disposed on the inner surface **123b** of the camera housing **120**. According to an embodiment, the damper **300** may be formed on the inner surface **123b** of the camera housing **120** through an injection-molding process.

[0085] In a state in which the electronic device (e.g., the electronic device **1001** of FIG. **10**) and/or the camera module **10** is not shaken (initial state), the frame **170** and the third carrier **160** may be formed to be spaced apart from each other by a third gap **g3**. The damper **300** of the disclosure may be formed so that the end portion **321** is spaced apart from the frame **170** by a fourth gap **g2** in a state in which the electronic device (e.g., the electronic device **1001** of FIG. **10**) and/or the camera module **10** is not shaken. When the electronic device is shaken and the frame **170** approaches the second carrier **150**, the damper **300** fixed to the camera housing **120** may also approach the frame **170**, and when the frame **170** further approaches the damper **300**, the damper **300** may contact the rear surface **172b** of the frame **170**. The damper **300** in contact with the rear surface **172b** of the frame **170** may be pressed by the frame **170** and bent in one direction as illustrated in FIG. **8** to be described below.

[0086] The damper **300** may at least partially include a portion **310** fixed to the camera housing and a portion **320** extending inclined from the optical axis. The extending portion **320** may be disposed in a space **S2** between the frame **170** and the second carrier **150**. According to an embodiment, the damper **300** may be disposed in the space **S2** between the fourth side wall **173** of the frame **170** and the second carrier **150**. The damper **300** may include a first portion **310** and a second portion **320** extending from the first portion **310** to be a single body. The first portion **310** may be a portion of

the damper **300** whose position and/or shape is fixed to the inner surface **123b** of the camera housing **120**, and the second portion **320** may be a portion whose position and/or shape is variable. According to an embodiment, the first portion **310** and the camera housing **120** may be formed through a double injection-molding process. According to an embodiment, the damper **300** may be guided by an opening **174** formed in the fourth side wall **173** of the frame **170**.

[0087] According to various embodiments, the category of the disclosure is not limited to the embodiments illustrated in FIGS. 5A and 5B, and/or FIGS. 6A and 6B. According to an embodiment, the embodiments illustrated in FIGS. 5A and 5B may be combined with the embodiments illustrated in FIGS. 6A and 6B. According to an embodiment of the disclosure, the electronic device (e.g., the electronic device **1001** of FIG. **10**) or the camera module **10** may include both the damper **200** illustrated in FIGS. 5A and 5B and the damper **300** illustrated in FIGS. 6A and 6B. For example, a plurality of dampers are provided at a position corresponding to one of the side walls of the camera module **10** (e.g., the 1-4th side wall **143d**). Referring to FIGS. 5A and 6A, the camera module **10** may include two dampers disposed symmetrically to the left and right around the lens assembly **110**. In this case, the damper **200** according to the embodiment illustrated in FIGS. 5A and 5B may be applied as one damper, and the damper **300** according to the embodiment illustrated in FIGS. 6A and 6B may be applied as another damper.

[0088] Hereinafter, the first portion **210** and the second portion **220** of the damper **200** are described in more detail with reference to FIGS. 7 and 8. The description of the damper **200** of FIGS. 7 and 8 may be applied to the damper **300** described above in FIGS. 6A and 6B.

[0089] FIG. 7 is a front view illustrating a damper **200** according to an embodiment of the disclosure.

[0090] FIG. 8 is a view illustrating a state in which the damper **200** of FIG. 7 is bent when contacting a cover plate **122** of a camera housing **120** according to an embodiment of the disclosure.

[0091] FIGS. 7 and 8 illustrate a state in which the damper **200** is disposed at a position corresponding to the 1-4th side wall **144c** except for the 1-1th side wall **143a**, the 1-2th side wall **143b**, and the 1-3th side wall **143c** of the first carrier **140**.

[0092] Referring to FIGS. 5A, 5B, 6A, 6B, 7, and 8 together, the first portion **210** of the damper **200** may include a first end portion **211**, and the second portion **220** may include a second end portion **221**. According to an embodiment, the first end portion **211** may be formed as a fixed end, and the second end portion **221** may be formed as a free end. According to an embodiment, the first portion **210** of the damper **200** may be fixed to the frame **170**, and the second portion **220** may have a shape extending toward the camera housing **120** (more specifically, the cover plate **122** of the camera housing **120**). When the second portion **220** contacts the camera housing **120**, it may be bent in one direction as illustrated in FIG. 8. According to an embodiment, the damper **200** may primarily absorb and/or disperse the impact when it contacts the camera housing **120**, and as the damper **200** moves slightly further toward the camera housing **120**, the second portion **220** may be bent in one direction to secondarily absorb and/or disperse the impact. According to an embodiment, the damper **200** is bent in one direction when contacting the camera housing **120**, and if the contact with the camera housing **120** is released, the shape may be restored, and its position may also be restored to face in the direction inclined from the optical axis O-I.

[0093] The damper **200** may be flexibly changed in shape. According to an embodiment, the damper **200** may include a material having elasticity such as rubber and urethane. Therefore, even if a portion of the damper **200** (e.g., the second portion **220**) is pressed and deformed, it may be restored to its original shape when the pressing is released.

[0094] Referring to FIG. 8, a direction where the damper **200** is bent may be a direction parallel to the X-axis when viewed from the second end portion **221** of the second portion **220**. Alternatively, when looking at the second end **221** of the second portion **220** from the point C, the direction in which the damper **200** is bent may correspond to the clockwise direction. The direction in which

the damper **200** is bent may be different according to embodiments. The damper **200** may face in any direction as long as it is configured to bend only in one direction when pressed by the rear surface **122b** of the cover plate **122** of the camera housing **120**.

[0095] Referring back to FIG. 7, the first portion **210** may be fixed to the frame **170** at the point C, and in this case, the point C may be the position of the center of the first end portion **211**. The second portion **220** may be elongated from the first portion **210** so that the second end portion **221** faces the camera housing **120**. In this case, the second portion **220** may not extend in parallel with the optical axis O-I, but may extend in a direction inclined from the optical axis O-I. According to an embodiment, the second end portion **221** of the second portion **220** may be formed at a position extending along an imaginary line drawn from the point C of the first portion **210** to be inclined at a predetermined angle from the optical axis O-I.

[0096] According to an embodiment, the second portion **220** may be described in more detail with reference to the imaginary line A-A' drawn parallel to the optical axis O-I with respect to the point C and the imaginary line B-B' drawn perpendicular thereto. According to an embodiment, the second portion **220** may have a length 'L' of the portion extending along the imaginary line drawn from the point C toward the second end portion **221** of the damper **200**. When the lengths of the portions corresponding to the length L on the imaginary line A-A' and the imaginary line B-B' are 'a' and 'b', respectively, the length 'L' of the second portion **220** may be " $\sqrt{a^2+b^2}$ ". For example, when the gap between the damper **200** and the camera housing **120** is set to be spaced apart by the second gap g2, and it is disposed on the imaginary line A-A' drawn parallel to the optical axis O-I, the damper **200** has a length of 'a', whereas, if it is disposed in a direction inclined from the optical axis O-I, the damper **200** may have a length of " $\sqrt{a^2+b^2}$ ", larger than the length 'a'. When the damper **200** is disposed in the direction inclined from the optical axis O-I, the length in which the damper **200** (more specifically, the second portion **220** of the damper **200**) may be bent when pressed by the rear surface **122b** of the cover plate **122** of the camera housing **120** may be increased as compared with when the damper **200** is disposed on the imaginary line A-A' drawn parallel to the optical axis O-I.

[0097] As in the embodiment of FIGS. 7 and 8, as the damper **200** is disposed to face in the direction inclined from the optical axis O-I, the damper **200** may be bent, increasing the effect of secondarily absorbing and/or dispersing impact as the damper **200** is bent.

[0098] FIG. 9A is a front view illustrating a damper according to an embodiment of the disclosure.

[0099] FIG. 9B is a view illustrating a state in which the damper of FIG. 9A is bent when contacting a cover plate of a camera housing according to an embodiment of the disclosure.

[0100] The shape of the damper is not limited to a certain specific shape. For example, as illustrated in FIG. 7, at least a portion of the cross-sectional shape of the damper **200** may have a wedge shape before deformation, but other shapes may also be applied.

[0101] Referring to FIGS. 9A and 9B, a damper **400** having a cross section at least partially in a zigzagged shape may be applied as well. The damper **400** illustrated in FIGS. 9A and 9B may also include a first portion **410** and a second portion **420**. The first portion **410** may include a first end portion forming a fixed end, and the second portion **420** may include a second end portion **421** forming a free end. Similar to the damper **200** of FIGS. 7 and 8, the damper **400** may also have a length 'L' of the portion extending along the imaginary line drawn from the point C toward the second end portion **421** of the damper **400**. According to an embodiment, similar to the damper **200** of FIGS. 7 and 8, the damper **400** may also be formed so that the imaginary line drawn from the point C toward the second end portion **421** of the damper **400** is inclined from the optical axis O-I.

[0102] As the dampers **200**, **300**, and **400** of the disclosure, any damper which has a shape in which the length is large as compared with its cross-sectional width, and may primarily absorb and/or disperse impact when contacting the camera housing **120** and, as its own shape bends or swells, secondarily absorb and/or disperse the impact may be included in the category of the disclosure.

[0103] FIG. 10 is a block diagram illustrating an electronic device in a network environment

according to an embodiment of the disclosure.

[0104] Referring to FIG. 10, the electronic device **1001** in the network environment **1000** may communicate with at least one of an electronic device **1002** via a first network **1098** (e.g., a short-range wireless communication network), or an electronic device **1004** or a server **1008** via a second network **1099** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **1001** may communicate with the electronic device **1004** via the server **1008**. According to an embodiment, the electronic device **1001** may include a processor **1020**, memory **1030**, an input module **1050**, a sound output module **1055**, a display module **1060**, an audio module **1070**, a sensor module **1076**, an interface **1077**, a connecting terminal **1078**, a haptic module **1079**, a camera module **1080**, a power management module **1088**, a battery **1089**, a communication module **1090**, a subscriber identification module (SIM) **1096**, or an antenna module **1097**. In an embodiment, at least one (e.g., the connecting terminal **1078**) of the components may be omitted from the electronic device **1001**, or one or more other components may be added in the electronic device **101**. According to an embodiment, some (e.g., the sensor module **1076**, the camera module **1080**, or the antenna module **1097**) of the components may be integrated into a single component (e.g., the display module **1060**).

[0105] The processor **1020** may execute, for example, software (e.g., a program **1040**) to control at least one other component (e.g., a hardware or software component) of the electronic device **1001** coupled with the processor **1020**, and may perform various data processing or computation.

According to one embodiment, as at least part of the data processing or computation, the processor **1020** may store a command or data received from another component (e.g., the sensor module **1076** or the communication module **1090**) in volatile memory **1032**, process the command or the data stored in the volatile memory **1032**, and store resulting data in non-volatile memory **1034**.

According to an embodiment, the processor **1020** may include a main processor **1021** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **1023** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **1021**. For example, when the electronic device **1001** includes the main processor **1021** and the auxiliary processor **1023**, the auxiliary processor **1023** may be configured to use lower power than the main processor **1021** or to be specified for a designated function. The auxiliary processor **1023** may be implemented as separate from, or as part of the main processor **1021**.

[0106] The auxiliary processor **1023** may control at least some of functions or states related to at least one component (e.g., the display module **1060**, the sensor module **1076**, or the communication module **1090**) among the components of the electronic device **1001**, instead of the main processor **1021** while the main processor **1021** is in an inactive (e.g., sleep) state, or together with the main processor **1021** while the main processor **1021** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **1023** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **1080** or the communication module **1090**) functionally related to the auxiliary processor **1023**. According to an embodiment, the auxiliary processor **1023** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. The artificial intelligence model may be generated via machine learning. Such learning may be performed, e.g., by the electronic device **1001** where the artificial intelligence is performed or via a separate server (e.g., the server **1008**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a

combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0107] The memory **1030** may store various data used by at least one component (e.g., the processor **1020** or the sensor module **1076**) of the electronic device **1001**. The various data may include, for example, software (e.g., the program **1040**) and input data or output data for a command related thereto. The memory **1030** may include the volatile memory **1032** or the non-volatile memory **1034**.

[0108] The program **1040** may be stored in the memory **1030** as software, and may include, for example, an operating system (OS) **1042**, middleware **1044**, or an application **1046**.

[0109] The input module **1050** may receive a command or data to be used by other component (e.g., the processor **1020**) of the electronic device **1001**, from the outside (e.g., a user) of the electronic device **1001**. The input module **1050** may include, for example, a microphone, a mouse, a keyboard, keys (e.g., buttons), or a digital pen (e.g., a stylus pen).

[0110] The sound output module **1055** may output sound signals to the outside of the electronic device **1001**. The sound output module **1055** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0111] The display module **1060** may visually provide information to the outside (e.g., a user) of the electronic device **1001**. The display **1060** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display **1060** may include a touch sensor configured to detect a touch, or a pressure sensor configured to measure the intensity of a force generated by the touch.

[0112] The audio module **1070** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **1070** may obtain the sound via the input module **1050**, or output the sound via the sound output module **1055** or a headphone of an external electronic device (e.g., the electronic device **1002**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **1001**.

[0113] The sensor module **1076** may detect an operational state (e.g., power or temperature) of the electronic device **1001** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **1076** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an accelerometer, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0114] The interface **1077** may support one or more specified protocols to be used for the electronic device **1001** to be coupled with the external electronic device (e.g., the electronic device **1002**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **1077** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0115] A connecting terminal **1078** may include a connector via which the electronic device **1001** may be physically connected with the external electronic device (e.g., the electronic device **1002**). According to an embodiment, the connecting terminal **1078** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0116] The haptic module **1079** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or motion) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **1079** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0117] The camera module **1080** may capture a still image or moving images. According to an embodiment, the camera module **1080** may include one or more lenses, image sensors, image signal processors, or flashes.

[0118] The power management module **1088** may manage power supplied to the electronic device **1001**. According to an embodiment, the power management module **1088** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0119] The battery **1089** may supply power to at least one component of the electronic device **1001**. According to an embodiment, the battery **1089** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0120] The communication module **1090** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **1001** and the external electronic device (e.g., the electronic device **1002**, the electronic device **1004**, or the server **1008**) and performing communication via the established communication channel. The communication module **1090** may include one or more communication processors that are operable independently from the processor **1020** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **1090** may include a wireless communication module **1092** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **1094** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via a first network **1098** (e.g., a short-range communication network, such as Bluetooth™ wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network **1099** (e.g., a long-range communication network, such as a legacy cellular network, a fifth generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., local area network (LAN) or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **1092** may identify or authenticate the electronic device **1001** in a communication network, such as the first network **1098** or the second network **1099**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **1096**.

[0121] The wireless communication module **1092** may support a 5G network, after a fourth generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **1092** may support a high-frequency band (e.g., the millimeter wave (mmWave) band) to achieve, e.g., a high data transmission rate. The wireless communication module **1092** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **1092** may support various requirements specified in the electronic device **1001**, an external electronic device (e.g., the electronic device **1004**), or a network system (e.g., the second network **1099**). According to an embodiment, the wireless communication module **1092** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0122] The antenna module **1097** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device). According to an embodiment, the antenna module may

include an antenna including a radiator formed of a conductor or conductive pattern formed on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **1097** may include a plurality of antennas (e.g., an antenna array). In this case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **1098** or the second network **1099**, may be selected from the plurality of antennas by, e.g., the communication module **1090**. The signal or the power may then be transmitted or received between the communication module **1090** and the external electronic device via the selected at least one antenna. According to an embodiment, other parts (e.g., radio frequency integrated circuit (RFIC)) than the radiator may be further formed as part of the antenna module **1097**.

[0123] According to an embodiment, the antenna module **1097** may form a mmWave antenna module. According to an embodiment, the mm Wave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0124] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0125] According to an embodiment, instructions or data may be transmitted or received between the electronic device **1001** and the external electronic device **1004** via the server **1008** coupled with the second network **1099**. The external electronic devices **1002** or **1004** each may be a device of the same or a different type from the electronic device **1001**. According to an embodiment, all or some of operations to be executed at the electronic device **1001** may be executed at one or more of the external electronic devices **1002** or **1004**, or the server **1008**. For example, if the electronic device **1001** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **1001**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **1001**. The electronic device **1001** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **1001** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **1004** may include an internet-of-things (IoT) device. The server **1008** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **1004** or the server **1008** may be included in the second network **1099**. The electronic device **1001** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0126] The electronic device according to embodiment(s) of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0127] An embodiment(s) of the disclosure and terms used therein are not intended to limit the technical features described in the disclosure to specific embodiments, and should be understood to include various modifications, equivalents, or substitutes of the embodiment. With regard to the

description of the drawings, similar reference numerals may be used to refer to similar or related elements. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0128] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0129] Various embodiments as set forth herein may be implemented as software (e.g., the program) including one or more instructions that are stored in a storage medium (e.g., internal memory or external memory) that is readable by a machine (e.g., the electronic device). For example, a processor (e.g., the processor) of the machine (e.g., the electronic device) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0130] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0131] According to embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. Some of the plurality of entities may be separately disposed in different components. According to embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0132] According to an embodiment of the disclosure, there may be provided an electronic device **1001** including a camera module **10**. The camera module **10** may include a camera housing **120**, a lens assembly **110** including at least one lens aligned along an optical axis O-I, a focus adjustment driver (AF driver) **M1** configured to move the lens assembly in a direction of the optical axis, an image stabilization driver (OIS driver) **M2**, **M3** configured to move the lens assembly in at least one direction crossing the optical axis, a frame **170** disposed in the camera housing and at least partially surrounding the focus adjustment driver **M1** and/or the image stabilization driver **M2**, **M3**, and a damper **200**, **300** including one end portion fixed to the frame or the camera housing and another end portion having a shape elongated toward the camera housing and configured to be bent in one direction based on the damper contacting the camera housing.

[0133] According to an embodiment, the damper **200**, **300** may be disposed in a direction inclined from the optical axis.

[0134] According to an embodiment, the damper **200**, **300** may be configured to be bent in one direction based on the damper contacting the camera housing, and the damper is configured to return to an inclined orientation with respect to the optical axis based on disengagement from the camera housing.

[0135] According to an embodiment, the camera housing **120** may include a cover plate **122** including a front surface **122a** facing in a first direction and a rear surface **122b** facing in a second direction opposite to the first direction, and having an optical opening **121**, and a side member **123** including a side surface **123a** facing in a third direction substantially perpendicular to the first direction and the second direction and connected to the cover plate **122**, and form a space capable of accommodating the lens assembly using the cover plate **122** and the side member **123**.

[0136] According to an embodiment, a cross section of the camera module may be formed in a polygonal shape with more sides than a rectangle.

[0137] According to an embodiment, the damper **200**, **300** may be disposed at a position corresponding to a remaining side surface except for a side surface corresponding to a position the image stabilization driver **M2**, **M3** and a side surface corresponding to a position of the focus adjustment driver **M1** of the camera module.

[0138] According to an embodiment, the frame **170** may include a first surface **172a** facing the cover plate, and a second surface **173a** facing the side member, and the damper **200** may be disposed on the second surface **173a** of the frame.

[0139] According to an embodiment, the damper **200** may be configured to be spaced apart from the camera housing **120** by a predetermined distance (second gap **g2**) in an initial state of not contacting the camera housing **120**.

[0140] According to an embodiment, the damper **300** may be disposed on an inner surface **123b** of the camera housing **120**.

[0141] According to an embodiment, the damper **300** may be configured to be spaced apart from the frame **170** by a predetermined distance (fourth gap **g4**) in an initial state of not contacting the frame **170**.

[0142] According to an embodiment, the camera module **10** further may include an image sensor **130** configured to form an image from light passing through the at least one lens, and a PCB **131** on which the image sensor may be disposed.

[0143] According to an embodiment, the camera module **10** further may include a first carrier **140** configured to be fixedly disposed on the PCB **131**.

[0144] According to an embodiment, the camera module **10** further may include a second carrier **150** overlapping the first carrier **140** and configured to be reciprocated in the optical axis direction.

[0145] According to an embodiment, the camera module **10** further may include a third carrier **160** overlapping the second carrier **150** and configured to be reciprocated in a direction crossing the optical axis.

[0146] According to an embodiment, the frame **170** may be fixedly coupled with the second carrier

150.

[0147] According to an embodiment, the damper **200** may at least partially include a portion **210** fixed to the frame and a portion **220** extending inclined from the optical axis and may be disposed in a space **S1** between the second carrier **150** and the camera housing **120**.

[0148] According to an embodiment, the damper **300** may at least partially include a portion **310** fixed to the camera housing and a portion **320** extending inclined with respect to the optical axis. The extending portion **320** may be disposed in a space **S2** between the first carrier **140** and the second carrier **150**.

[0149] According to an embodiment, the camera module **10** may include a focus adjustment driver and an image stabilization driver of a lens shift type.

[0150] According to an embodiment, the camera module **10** may include a focus adjustment driver and an image stabilization driver of an image sensor shift type.

[0151] According to an embodiment of the disclosure, there may be provided a camera module comprising a camera housing **120**, a lens assembly including at least one lens aligned along an optical axis O-I, a carrier member configured to guide the lens assembly in an optical-axis direction and/or a direction crossing the optical axis, and a driving member **M1**, **M2**, **M3** including at least one coil, and at least one magnet and a plurality of guide balls disposed to at least partially face the at least one coil, and moving the carrier member in the optical-axis direction and the direction crossing the optical axis in a ball bearing manner, a frame **170** disposed in the camera housing and at least partially surrounding the driving member **M1**, **M2**, **M3**, and a damper **200**, **300** having one end portion fixed to the frame and another end portion having a shape elongated toward the camera housing and configured to be bent in one direction based on the damper contacting the camera housing.

[0152] According to an embodiment, the camera module **10** further may include an image sensor **130** configured to form an image from light passing through the at least one lens, and a PCB **131** on which the image sensor may be disposed.

[0153] According to an embodiment, the carrier member includes a first carrier **140** coupled with the first PCB and including a first opening **141** formed at a position overlapping the image sensor and a first plate **142** at least partially surrounding the first opening, and including a first side wall **143** protruding from the first plate in the optical-axis direction, a second carrier **150** disposed on the first carrier and including a second opening **151** formed at a position corresponding to the first opening and a second plate **152** at least partially surrounding the second opening, and including a second side wall protruding from the second plate in the optical-axis direction, and a third carrier **160** disposed on the second carrier.

[0154] According to an embodiment, the at least one driving member **M1**, **M2**, **M3** may include a first driver **M1** including a first coil **144** disposed on a 1-1th side wall **143a** of the first side wall **143** and a first magnet **154** disposed at a position corresponding to the first coil on the second side wall **153** of the second carrier **150**, a second driver **M2** including a second coil **145** disposed on a 1-2th side wall **143b** of the first side wall **143** and a second magnet **165** disposed at a position corresponding to the second coil, around the third carrier, and a third driver **M3** including a third coil **146** disposed on a 1-3th side wall **143c** of the first side wall **143** and a third magnet **166** disposed at a position corresponding to the second coil, around the third carrier.

[0155] According to an embodiment, the damper **200**, **300** may be positioned at a position corresponding to a 1-4th side wall **143d** of the first side wall **143**.

[0156] While the disclosure has been described and shown in connection with various embodiments, it should be appreciated that various embodiments are intended as limiting the disclosure but as illustrative. It will be apparent to one of ordinary skill in the art that various changes may be made in form and detail without departing from the overall scope of the disclosure, including the appended claims and their equivalents. In the embodiments of FIGS. **1** to **4**, **5A**, **5B**, **6A**, **6B**, **7**, **8**, **9A**, and **9B**, it has been described that the first driver **M1**, the second driver **M2**, and

the third driver M3, respectively, are disposed on three different surfaces in the camera module **10** having a substantially rectangular cross section, and the auto-focus (AF) operation and image stabilization (OIS) operation are implemented based thereupon, but the disclosure is not limited thereto. For example, the camera module **10** may have a cross section in a shape (e.g., a polygon with five or more sides) other than the rectangle to have five or more surfaces and, accordingly, the positions of the first driver M1, the second driver M2, and the third driver M3 may be various. As another example, the shape of the damper **200** in the embodiments of FIGS. **1** to **4**, **5A**, **5B**, **6A**, **6B**, **7**, **8**, **9A**, and **9B** described above may be varied according to embodiments.

[0157] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device including: a camera, wherein the camera includes: a camera housing, a lens assembly including at least one lens aligned along an optical axis, a focus adjustment (AF) driver configured to move the lens assembly in a direction of the optical axis, an image stabilization (OIS) driver configured to move the lens assembly in at least one direction crossing the optical axis, a frame disposed in the camera housing and at least partially surrounding at least one of the AF driver or the image stabilization driver, and a damper including one end portion fixed to the frame or the camera housing and another end portion having a shape elongated toward the camera housing and configured to be bent in one direction based on the damper contacting the camera housing.
2. The electronic device of claim 1, wherein the damper is disposed in a direction inclined from the optical axis.
3. The electronic device of claim 2, wherein the damper is configured to be bent in one direction based on the damper contacting the camera housing, and wherein the damper is configured to return to an inclined orientation with respect to the optical axis based on disengagement from the camera housing.
4. The electronic device of claim 1, wherein the camera housing includes: a cover plate including a front surface facing in a first direction and a rear surface facing in a second direction opposite to the first direction, and having an optical opening, and a side member including a side surface facing in a third direction substantially perpendicular to the first direction and the second direction and connected to the cover plate, and wherein a space capable of accommodating the lens assembly is formed using the cover plate and the side member.
5. The electronic device of claim 4, wherein the damper is disposed at a position corresponding to a remaining side surface except for a side surface corresponding to a position the OIS driver and a side surface corresponding to a position of the AF driver of the camera among side surfaces of the side member.
6. The electronic device of claim 4, wherein the frame includes: a first surface facing the cover plate, and a second surface facing the side member, and wherein the damper is disposed on the second surface of the frame.
7. The electronic device of claim 1, wherein the damper is configured to be spaced apart from the camera housing by a predetermined distance in an initial state of not contacting the camera housing.
8. The electronic device of claim 5, wherein the damper is disposed on an inner surface of the camera housing.
9. The electronic device of claim 8, wherein the damper is configured to be spaced apart from the frame by a predetermined distance in an initial state of not contacting the frame.

- 10.** The electronic device of claim 1, wherein the camera further includes: an image sensor configured to form an image from light passing through the at least one lens; and a printed circuit board (PCB) on which the image sensor is disposed.
- 11.** The electronic device of claim 10, wherein the camera further includes a first carrier configured to be fixedly disposed on the PCB.
- 12.** The electronic device of claim 11, wherein the camera further includes a second carrier overlapping the first carrier and configured to be reciprocated in an optical axis direction.
- 13.** The electronic device of claim 12, wherein the camera further includes a third carrier overlapping the second carrier and configured to be reciprocated in a direction crossing the optical axis.
- 14.** The electronic device of claim 12, wherein the damper at least partially includes a portion fixed to the frame and a portion extending inclined from the optical axis, and wherein the extending portion is disposed in a space between the second carrier and the camera housing.
- 15.** The electronic device of claim 12, wherein the damper at least partially includes a portion fixed to the camera housing and a portion extending inclined with respect to the optical axis, and wherein the extending portion is disposed in a space between the frame and the second carrier.
- 16.** A camera comprising: a camera housing; a lens assembly including at least one lens aligned along an optical axis; a carrier member configured to guide the lens assembly in at least one of an optical-axis direction or a direction crossing the optical axis; and a driving member including: at least one coil, and at least one magnet and a plurality of guide balls disposed to at least partially face the at least one coil, and moving the carrier member in the optical-axis direction and the direction crossing the optical axis in a ball bearing manner, wherein a frame is disposed in the camera housing and at least partially surrounding the driving member, and wherein a damper having one end portion fixed to the frame or the camera housing and another end portion having a shape elongated toward the camera housing and is configured to be bent in one direction based on the damper contacting the camera housing.
- 17.** The camera of claim 16, wherein the camera includes an image sensor configured to form an image from light passing through the at least one lens, and a PCB on which the image sensor is disposed.
- 18.** The camera of claim 17, wherein the carrier member includes a first carrier coupled with the PCB and including a first opening formed at a position overlapping the image sensor and a first plate at least partially surrounding the first opening, and including a first side wall protruding from the first plate in the optical-axis direction, a second carrier disposed on the first carrier and including a second opening formed at a position corresponding to the first opening and a second plate at least partially surrounding the second opening, and including a second side wall protruding from the second plate in the optical-axis direction, and a third carrier disposed on the second carrier.
- 19.** The camera of claim 18, wherein the at least one driving member includes a first driver including a first coil disposed on a 1-1th side wall of the first side wall and a first magnet disposed at a position corresponding to the first coil on the second side wall of the second carrier, a second driver including a second coil disposed on a 1-2th side wall of the first side wall and a second magnet disposed at a position corresponding to the second coil, around the third carrier, and a third driver including a third coil disposed on a 1-3th side wall of the first side wall and a third magnet disposed at a position corresponding to the second coil, around the third carrier.
- 20.** The camera of claim 16, wherein the damper is positioned at a position corresponding to a 1-4th side wall of a first side wall.
-